# THE SMALL AND BIG ZARISKI, ETALE, SMOOTH, SYNTOMIC AND FPPF SITES HAVE ENOUGH POINTS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument for the whole list of sites, the affine and big versions being handled by the comparison of topologies inside the counted span.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- Having enough points for a site. Sites, the notion of a site with enough points, i.e. of a conservative family of fibre functors given by morphisms from the point. Supplies the property asserted for the fourteen sites listed.
- Comparison of the topologies and geometric points of a scheme. Same chapter, the description of points of the etale, smooth, syntomic and fppf topologies by geometric points of the scheme, and the standard facts that the affine versions and the big versions have the same points. Supplies the argument that the listed sites have the same conservative families of points.

*Source reference.* Original lemma 020W: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3824–3824. Navigation pointer invoked at proof line 3824; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 021E: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3825–3825. Navigation pointer invoked at proof line 3825; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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*Date*: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 06VC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3826–3826. Navigation pointer invoked at proof line 3826; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 06VD: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3827–3827. Navigation pointer invoked at proof line 3827; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 021V: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3828–3828. Navigation pointer invoked at proof line 3828; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0DW0: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3831–3831. Navigation pointer invoked at proof line 3831; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 3837; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 04HR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3838–3838. Navigation pointer invoked at proof line 3838; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $S$ be a scheme. All of the following sites have enough points $S_{affine,Zar}$, $S_{Zar}$, $S_{affine,étale}$, $S_{étale}$, $(Sch/S)_{Zar}$, $(Aff/S)_{Zar}$, $(Sch/S)_{étale}$, $(Aff/S)_{étale}$, $(Sch/S)_{smooth}$, $(Aff/S)_{smooth}$, $(Sch/S)_{syntomic}$, $(Aff/S)_{syntomic}$, $(Sch/S)_{fppf}$, and $(Aff/S)_{fppf}$.*

*Proof.* For each of the big sites the associated topos is equivalent to the topos defined by the site $(Aff/S)_\tau$, see Topologies, Lemmas 020W, 021E, 06VC, 06VD, and 021V. The result for the sites $(Aff/S)_\tau$ follows immediately from Deligne's result Sites, Lemma 0DW0.

The result for $S_{Zar}$ is clear. The result for $S_{affine,Zar}$ follows from Deligne's result. The result for $S_{étale}$ either follows from (the proof of) Theorem 03PU or from Topologies, Lemma 04HR and Deligne's result applied to $S_{affine,étale}$. □